Project - Low Level Design

on

Multi-Agent Education System

Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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1. Introduction

The **Multi-Agent Education System** is an intelligent application built using **Agentic AI architecture**, where multiple specialized agents collaborate to generate high-quality educational content. Traditional AI systems typically rely on a single model to process user queries, which may result in unstructured or incomplete responses. To address this limitation, this project introduces a **multi-agent collaborative framework** using CrewAI that enables task delegation, structured reasoning, and multimodal output generation.

The system includes multiple agents such as an **Orchestrator Agent**, **Content Generation Agent**, and **Image Generation Agent** that communicate through defined workflows. Each agent performs a specific function, improving the quality, modularity, and scalability of the system. The final output consists of well-structured educational explanations along with relevant images to enhance conceptual understanding.

1.1 Scope of the Document

This document describes the architecture and design of the **Multi-Agent Education System**. It explains how multiple AI agents collaborate to process educational queries and generate structured outputs.

The document includes:

- System architecture overview
- Description of agents and their responsibilities
- Workflow and process design
- Information exchange between agents
- API usage and integration
- Design considerations and scalability aspects

The scope of this document is limited to the conceptual design and implementation structure of the multi-agent educational system.

1.2 Intended Audience

This document is intended for:

Students and learners studying Agentic AI and multi-agent systems

Developers and engineers who want to implement AI-based educational systems

Project evaluators and instructors assessing the system design and implementation

Researchers interested in collaborative AI and intelligent agent architectures

1.3 System Overview

The Multi-Agent Education System consists of multiple AI agents working collaboratively under the control of an orchestrator.

Key Components:

Orchestrator (Crew Manager): Controls workflow and manages communication between agents

Researcher Agent: Collects relevant study material based on user input

Writer Agent: Converts raw data into structured and readable educational content

LLM API: Provides intelligence for generating responses

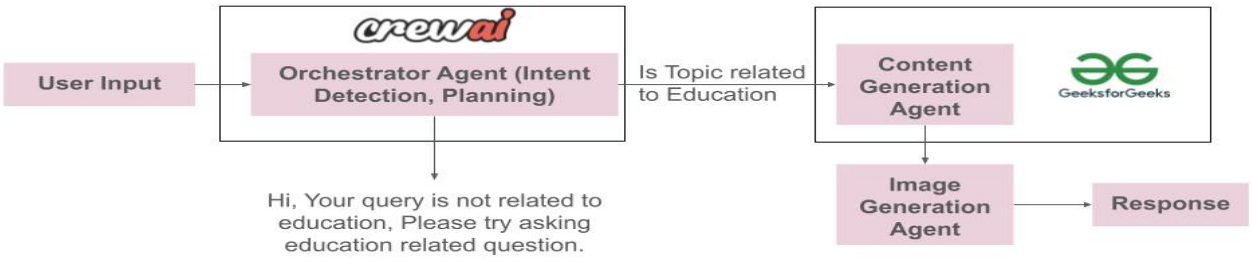
Image Generation Agent: Produces diagrams or visual learning aids.

Working Principle

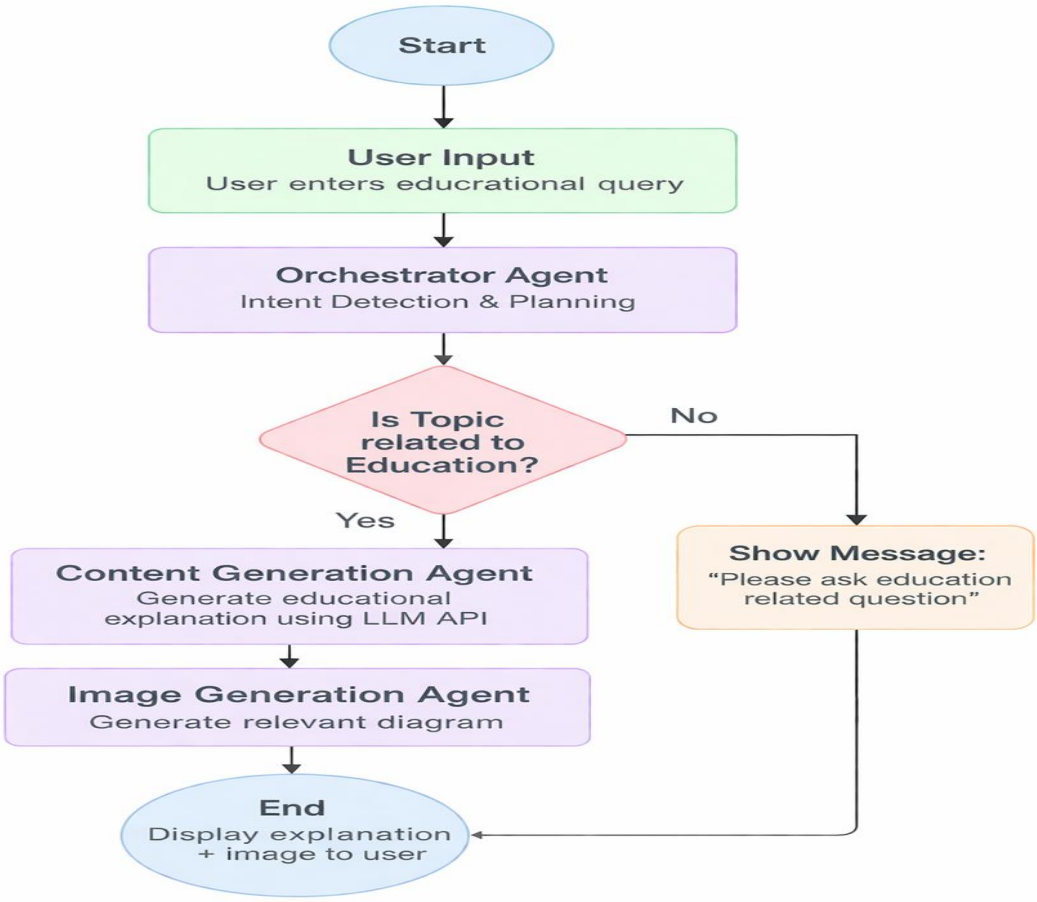
- User provides an educational query or topic through the interface.
- The Orchestrator Agent analyzes the query and checks whether it is related to education.
- If the query is not related to education, the system returns a message asking the user to enter an education-related question.
- If the query is valid, the Orchestrator Agent assigns the task to the Content Generation Agent.
- The Content Generation Agent generates structured educational content using the LLM API.
- The Image Generation Agent generates a relevant diagram or visual representation of the topic.
- The final response, including both explanation and image, is displayed to the user.

2 System Design

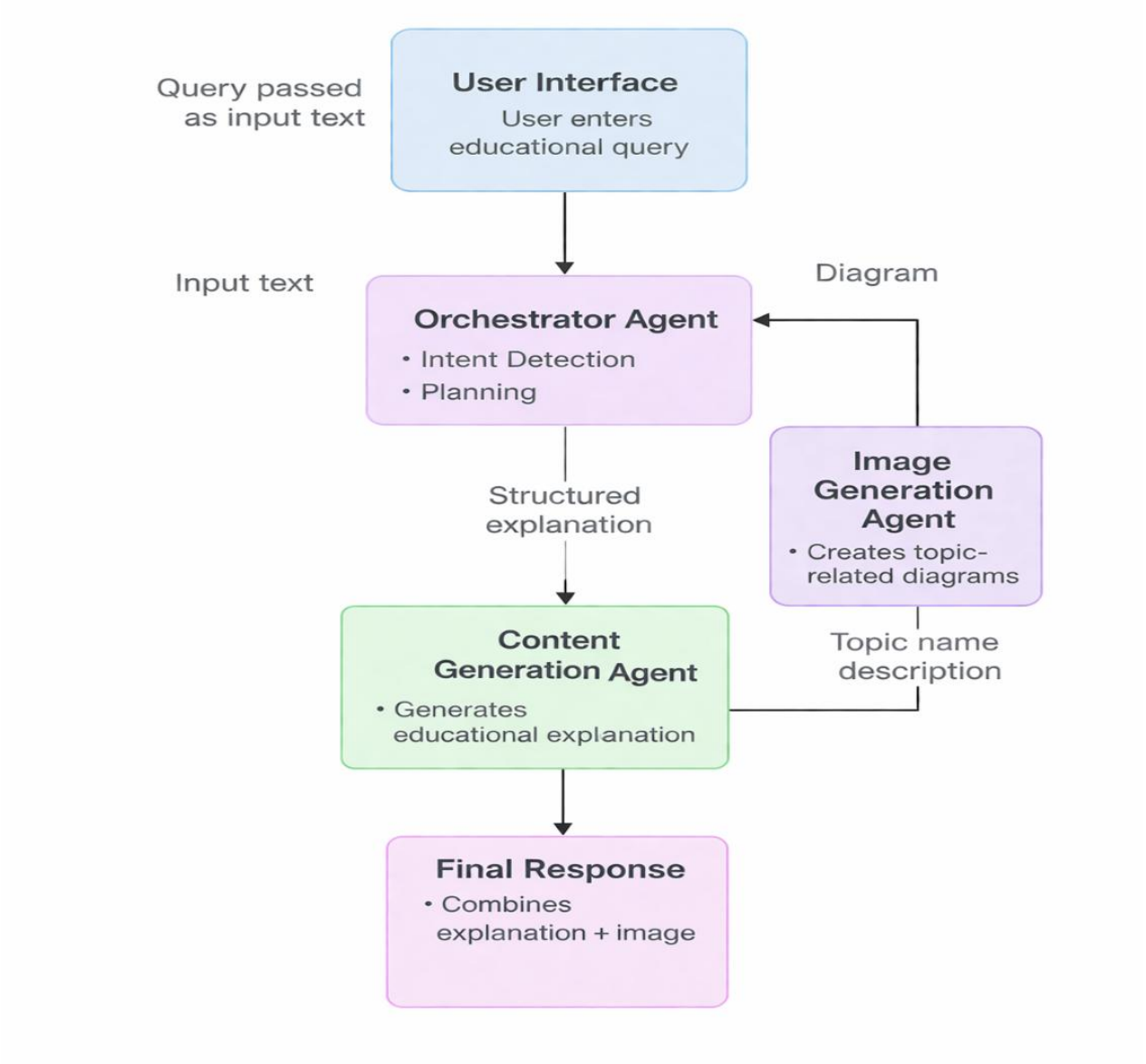
2.1 Application Design



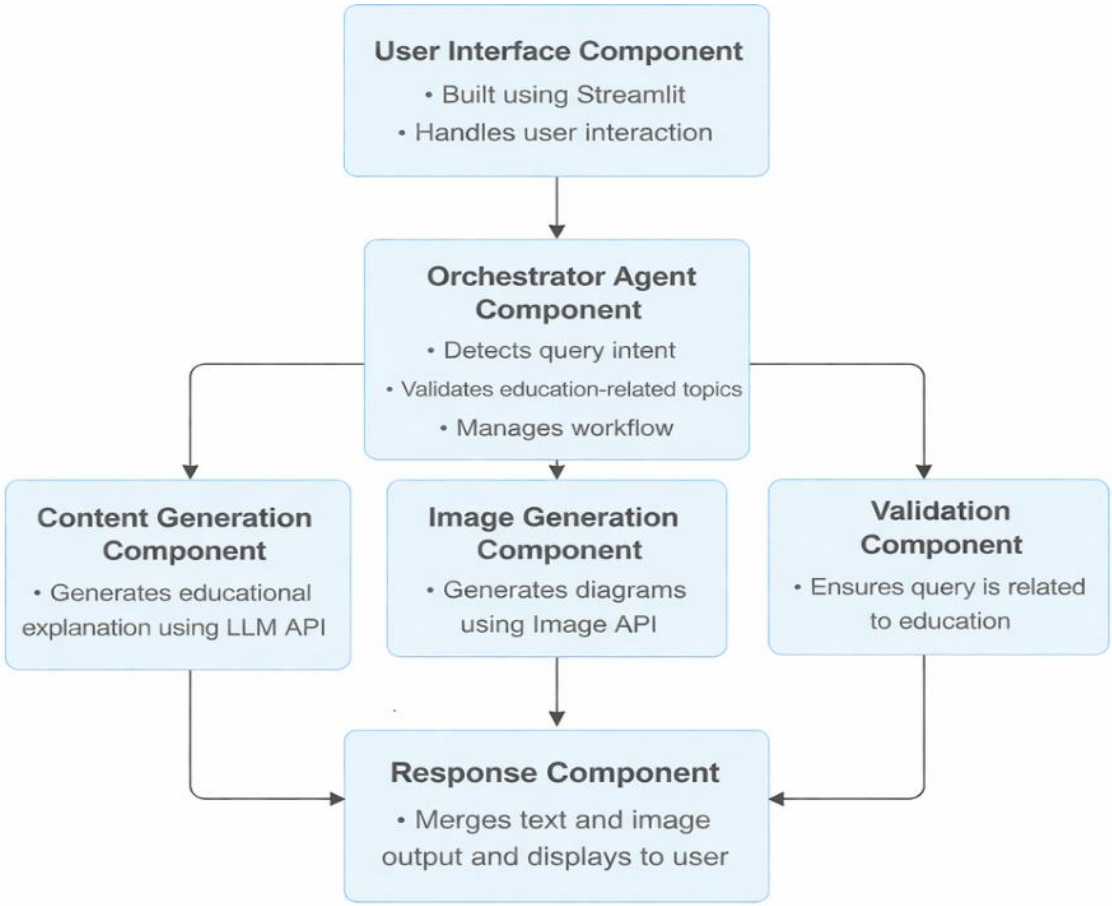
2.2 Process Flow



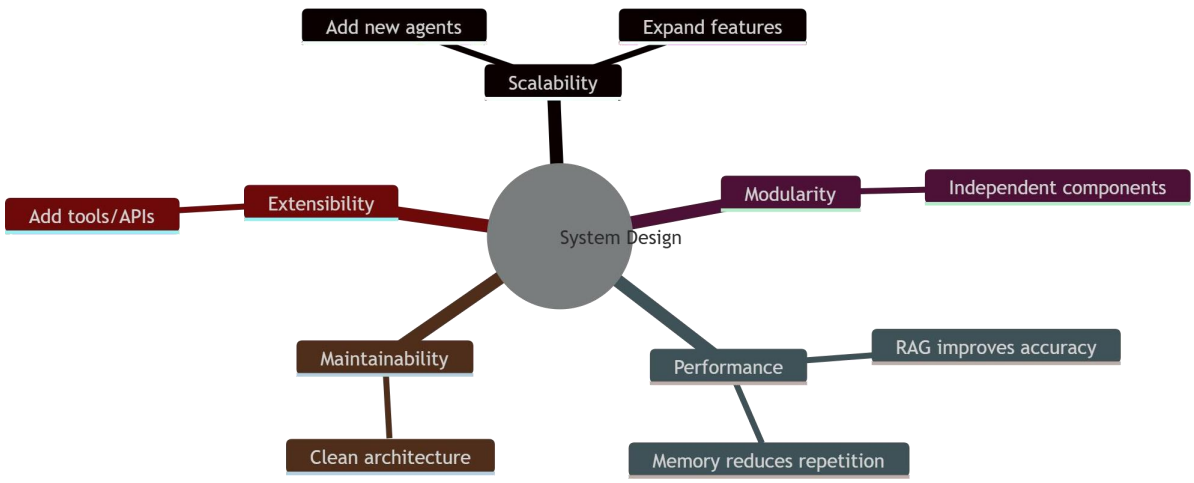
2.3 Information Flow



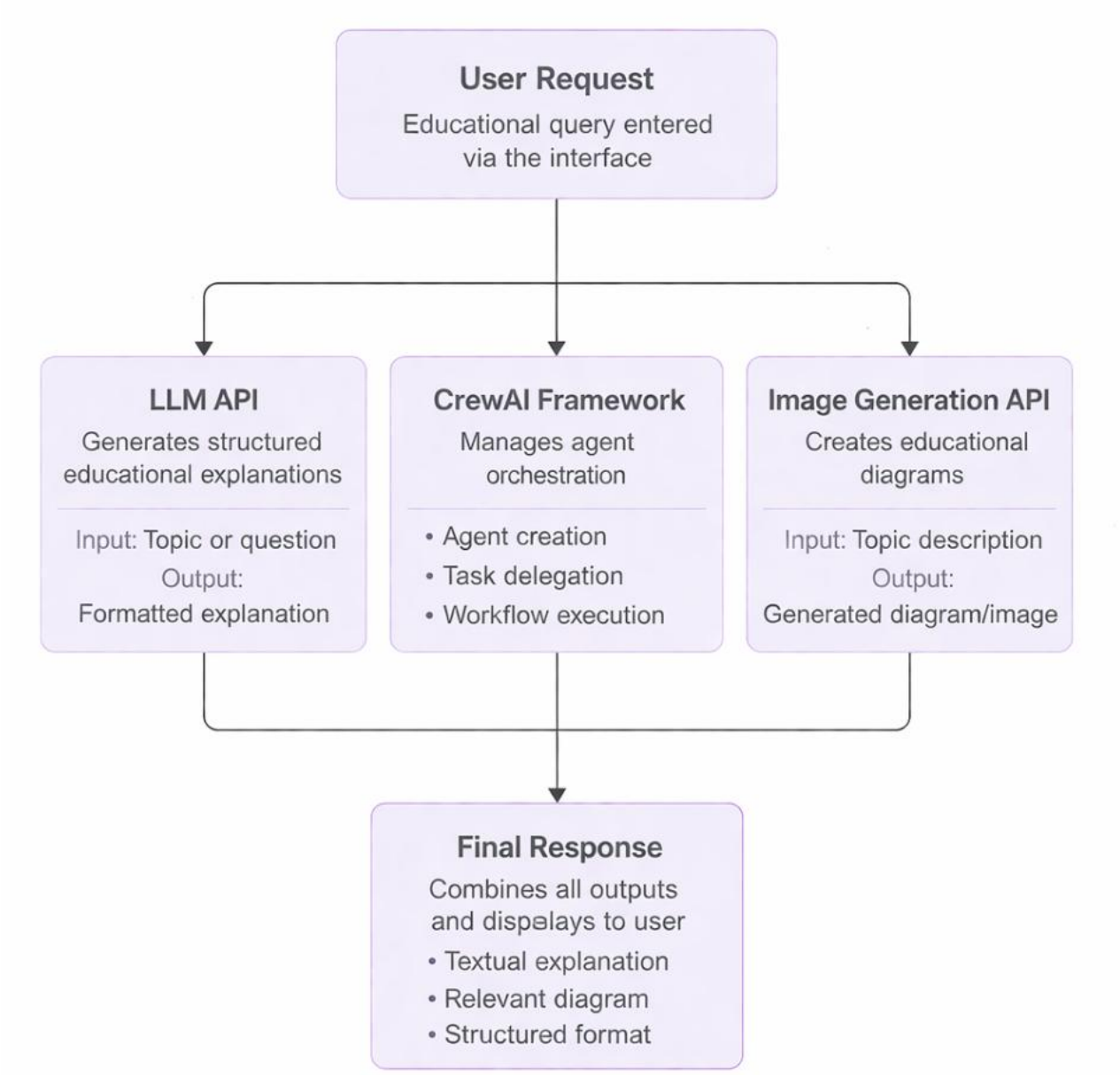
2.4 Components Design



2.5 Key Design Considerations



2.6 API Catalogue.



3. References

This project is based on concepts and technologies from the fields of **Agentic AI**, **Large Language Models (LLMs)**. Frameworks such as CrewAI and LangChain were used to design and manage the multi-agent workflow, while Python served as the primary implementation language. The system integrates LLM APIs to generate intelligent responses. Additional references include official documentation, research materials, and online resources related to multi-agent systems, AI orchestration, and modern NLP techniques, which collectively supported the design and development of this system.

Research Papers & Concepts

1. Russell, S., & Norvig, P.
   Artificial Intelligence: A Modern Approach
   Pearson Education
   Fundamental concepts of intelligent agents and AI systems.
2. OpenAI
   GPT Models and Generative AI Documentation
   Concepts of Large Language Models and text generation.
3. Lewis, P., et al.
   Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks
   (2020)
   Introduction to RAG architecture.